Products API Guide

Cosium

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Overview

HTTP verbs

RESTful Center tries to adhere as closely as possible to standard HTTP and REST conventions in its use of HTTP verbs.

Verb	Usage
`GET`	Used to retrieve a resource
`POST`	Used to create a new resource
`PUT`	Used to create or update a resource in an indempotent way
`PATCH`	Used to update an existing resource, including partial updates
`DELETE`	Used to delete an existing resource

HTTP status codes

RESTful Center tries to adhere as closely as possible to standard HTTP and REST conventions in its use of HTTP status codes.

Status code	Usage
`200 OK`	The request completed successfully
`201 Created`	A new resource has been created successfully. The resource's URI is available from the response's `Location` header
`204 No Content`	An update to an existing resource has been applied successfully
`400 Bad Request`	The request was malformed. The response body will include an error providing further information
`401 Unauthorized`	The request could not be authenticated. You must provide a valid access token.
`403 Forbidden`	The request has been authenticated but the operation is fobidden. This can be caused by insufficient privileges.
`404 Not Found`	The requested resource did not exist

Access token

Often, a valid access token is required to access this API resources.

How to get an access token

Request fields

Path	Type	Description
`login`	`String`	The login
`password`	`String`	The password
`site`	`String`	The site/store name to authenticate to

Response headers

Name	Description
`access_token`	The access token

Example request

HTTP

```http
POST /01ACJA05562/api/authenticate/basic HTTP/1.1
Content-Type: application/json
Content-Length: 123
Host: c1.cosium.biz

{
  "login" : "class com.cos.center.core.user.UserFixture59",
  "password" : "ReX^9ut4Kup>U)/E",
  "site" : "foo-site-42"
}
```

Example response

```http
HTTP/1.1 200 OK
Vary: Origin
Vary: Access-Control-Request-Method
Vary: Access-Control-Request-Headers
Set-Cookie: access_token=NNHiKn3rJrua3Xzs62s0cpl5z7U; Path=/01ACJA05562; Secure; HttpOnly
access_token: NNHiKn3rJrua3Xzs62s0cpl5z7U
COS-AUTH-TOKEN: NNHiKn3rJrua3Xzs62s0cpl5z7U
authentication-ok: true
Cache-Control: no-cache, no-store, max-age=0, must-revalidate
Pragma: no-cache
Expires: 0
Strict-Transport-Security: max-age=31536000 ; includeSubDomains
Content-Length: 27

NNHiKn3rJrua3Xzs62s0cpl5z7U
```

How to use the access token

You must pass the access token in each request either via:

- an Authorization AccessToken header. e.g. for token **abcd1234**, you must pass the header **Authorization: AccessToken abcd1234**
- a cookie named **access_token**
- a request parameter named **access_token**

Hypermedia

RESTful Cosium Center uses hypermedia and resources include links to other resources in their responses. Responses are in Hypertext Application from resource to resource. Language (HAL) (http://stateless.co/hal_specification.html) format. Links can be found beneath the `_links` key. Users of the API should not create URIs themselves, instead they should use the above-described links to navigate. To obtain HAL representation, you must provide the header **Accept: application/hal+json** to your Http request.

CORS - Cross-Origin Resource Sharing

CORS activation

CORS is disabled by default.

Web applications hosted on different domains need their domains to be added to CORS.

The Web Application CORS activation request must be made to a Cosium representative.

CORS cookie credentials

CORS enabled web applications relying on cookie credentials must set `XMLHttpRequest` parameter `withCredentials` to true.

Example with XMLHttpRequest

JAVASCRIPT

```javascript
var request = new XMLHttpRequest();
request.withCredentials = true;
```

MDN documentation (https://developer.mozilla.org/en-US/docs/Web/API/XMLHttpRequest/withCredentials)

Resources

Get the link for the Product API

Now that you are authenticated, you have to get the link of the product API. Make a GET Http request to the URI : http://c1.cosium.biz/01ACJA05562/api and then you will get the URI by following the link : products.

Products

Get the product filtered list

Example request

```http
GET /01ACJA05562/api/products?ean_code=109876543210&gtin_code=12345678910&code=999999999&family_type=OPT_FRAME HTTP/1.1
Accept: application/prs.hal-forms+json
Host: c1.cosium.biz
```

Request parameters

Parameter	Description
`ean_code`	EAN code or UPC code of a product
`gtin_code`	gtin code of a product
`code`	This is the product code
`family_type`	Family of the product

All parameters are optional, mind the cases, otherwise you'll receive an error. You can choose the size of your pagination result by using the parameter 'page_size'. Default page size is 10. You'll find the next and previous pages result by following the links 'previous-page' and 'next-page'. You'll find the first and the last pages result by following the linls 'first-page' and 'last-page'.

Example response

HTTP

```
HTTP/1.1 200 OK
Content-Type: application/prs.hal-forms+json
Content-Length: 1102

{
  "_embedded" : {
    "content" : [ {
      "productCode" : "999999999",
      "barcode" : "barcode",
      "gtinCode" : "00012345678910",
      "familyType" : "OPT_FRAME",
      "label" : "product label",
      "eanCode" : "109876543210",
      "stopProductionDate" : "2022-01-07T22:01:11.000Z",
      "_links" : {
        "self" : {
          "href" : "https://c1.cosium.biz/01ACJA05562/api/products/337"
        },
        "stock" : {
          "href" : "https://c1.cosium.biz/01ACJA05562/api/products/337/stock{?siteId}",
          "templated" : true
        },
        "latent-sales" : {
          "href" : "https://c1.cosium.biz/01ACJA05562/api/products/337/latent-sales{?quotations_max_age_in_days}",
          "templated" : true
        }
      }
    } ]
  },
  "_links" : {
    "self" : {
      "href" : "https://c1.cosium.biz/01ACJA05562/api/products?ean_code=109876543210&gtin_code=12345678910&code=999999999&f
    }
  },
  "page" : {
    "size" : 10,
    "totalElements" : 1,
    "totalPages" : 1,
    "number" : 0
  }
}
```

Response fields

Path	Type	Description
`_embedded.content[0].productCode`	`String`	Stock quantity
`_embedded.content[0].barcode`	`String`	Stock quantity
`_embedded.content[0].gtinCode`	`String`	Stock quantity
`_embedded.content[0].familyType`	`String`	Stock quantity
`_embedded.content[0].label`	`String`	Stock quantity
`_embedded.content[0].eanCode`	`String`	Stock quantity
`_embedded.content[0].stopProductionDate`	`String`	The stop production date
`page`	`Object`	Informations regarding the paging requested
`_embedded.content[0]._links`	`Object`	The product resources links
`_links`	`Object`	The hal links regarding a product

Product Stock

The result format can be different depending on the 'Accept' header of your Http request. If you put nothing, by default a JSON content will be returned. If you add the header 'Accept: application/hal+json', the content will be displayed as HAL structure.

Request parameters

Parameter	Description
`siteId`	Site id, defaults to current site

Example request

```http
GET /01ACJA05562/api/products/337/stock?siteId=1 HTTP/1.1
Accept: application/prs.hal-forms+json
Host: c1.cosium.biz
```

Response fields

Path	Type	Description
`quantity`	`Number`	Stock quantity
`_links`	`Object`	The hal links regarding a product

Example response

```http
HTTP/1.1 200 OK
Content-Type: application/prs.hal-forms+json
Content-Length: 274

{
  "quantity" : 13,
  "_links" : {
    "self" : {
      "href" : "https://c1.cosium.biz/01ACJA05562/api/products/337/stock{?siteId}",
      "templated" : true
    },
    "product" : {
      "href" : "https://c1.cosium.biz/01ACJA05562/api/products/337"
    }
  }
}
```

Product Latent Sales

Latent sales consist in the calculation of the quantity of a product that is implied in quotes but not transfered into invoices yet.

Request parameters

Parameter	Description
`quotations_max_age_in_days`	Number that indicates the maximum of days to look for

Example request

```http
GET /01ACJA05562/api/products/337/latent-sales?quotations_max_age_in_days=10 HTTP/1.1
Accept: application/prs.hal-forms+json
Host: c1.cosium.biz
```

The parameter 'quotations_max_age_in_days' is optional. Its value by default is 90 days.

Example response

```http
HTTP/1.1 200 OK
Content-Type: application/prs.hal-forms+json
Content-Length: 275

{
  "quantity" : 3,
  "_links" : {
    "self" : {
      "href" : "https://c1.cosium.biz/01ACJA05562/api/products/337/latent-sales?quotations_max_age_in_days=10"
    },
    "product" : {
      "href" : "https://c1.cosium.biz/01ACJA05562/api/products/337"
    }
  }
}
```

Response fields

Path	Type	Description
`quantity`	`Number`	Latent sales quantity
`_links`	`Object`	The hal links regarding a product

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